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2027 CYCLE

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CUET UG

Biology

Human Reproduction

Official Syllabus Unit VI · Class 12 · CUET UG 2027 Cycle Notes

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Syllabus Alignment

Sub-topic	Official syllabus pointer (NTA CUET(UG) Biology syllabus 2026, the current 2027 baseline, Unit VI)	Skills the questions test
Male and female reproductive system	Male and female reproductive system;	Naming every part and matching it to its one-line function.
Microscopic anatomy of testis and ovary	Microscopic anatomy of testis and ovary;	Identifying cell types in a section and stating what each one does.
Gametogenesis — spermatogenesis and oogenesis	Gametogenesis – spermatogenesis and oogenesis;	Ordering the stages and tracking chromosome numbers (46 vs 23).
Menstrual cycle	Menstrual cycle	Placing the four phases and the hormones on the 28-day timeline.
Fertilisation and embryo development up to blastocyst, implantation	Fertilisation, embryo development up to blastocyst formation, implantation;	Sequencing events from sperm entry to the buried blastocyst.
Pregnancy, embryonic development and placenta	Pregnancy and embryonic development and placenta;	Linking placenta functions and foetal milestones to the right month.
Parturition and lactation	Parturition and lactation	Explaining the oxytocin feedback loop and the value of colostrum.

EXAM PATTERN

CUET UG pattern (NTA official information bulletin and syllabus, 2026 cycle — the current 2027 baseline): each subject paper carries **50 compulsory questions** in **60 minutes**, with +5 for every correct answer and −1 for every incorrect answer. The Biology paper maps strictly to NCERT Class 12, and Unit VI opens the reproduction block: systems, gametes, the cycle that times them, and the nine months that follow.

The Male Reproductive System

The male system is a factory with a delivery route. The testes make the sperm, a ladder of tubes matures and stores them, and three glands blend their fluids into semen. Every part has exactly

one job, and questions love to hand you a part and ask for that job. So we will walk the whole line, from the factory floor to the final exit, and then pin a one-line function to every part.

- **The scrotum — the cooling pouch:** a bag of skin that hangs the two testes outside the body cavity.
- **Why outside?** Making sperm needs a temperature 2–2.5 °C **below** body temperature; the scrotum holds the testes in exactly that cooler zone.
- **The testes — paired gonads:** gonads are organs that make gametes (sex cells) plus sex hormones; each testis is an oval gland about 4–5 cm long.
- **Seminiferous tubules:** inside each testis are tightly coiled tubes called seminiferous tubules — the actual sperm factories.
- **Male germ cells (spermatogonia):** cells lining each tubule that divide and slowly turn into sperm; spermatogonium is the name for one such cell.
- **Sertoli cells — the nurse cells:** tall cells inside the tubule wall that feed, protect and finally release the growing sperm.
- **Interstitial (Leydig) cells:** they sit in the spaces **between** tubules and secrete testosterone, the hormone that drives sperm production and male traits.
- **The duct ladder, in order:** rete testis → vasa efferentia → epididymis → vas deferens → ejaculatory duct → urethra.
- **The epididymis:** a tightly coiled tube pressed against each testis; it stores sperm for weeks and matures them so they learn to swim.
- **Accessory glands:** seminal vesicles, prostate gland and bulbourethral (Cowper's) glands — their fluids mix with sperm to form semen.
- **Semen, the finished mix:** sperm are only a small share of semen; most of the volume is gland fluid that feeds and protects them on the journey.
- **Why alkaline matters:** the vagina is acidic, so the semen's alkaline secretions shield the sperm as they enter.

Every part, one line each

Part	Structure in one line	The job it does
Scrotum	Skin pouch outside the abdomen	Keeps testes 2–2.5 °C cooler for sperm making
Testis	Paired oval gonad packed with tubules	Makes sperm and testosterone
Seminiferous tubule	Coiled tube inside the testis	Site of sperm production
Sertoli cell	Nurse cell in the tubule wall	Feeds developing sperm; releases finished ones
Leydig cell	Sits in spaces between tubules	Secretes testosterone
Rete testis	Tiny network where tubules empty	Collects sperm from the tubules
Vasa efferentia	Short fine tubes	Carry sperm to the epididymis

Part	Structure in one line	The job it does
Epididymis	Coiled tube on the testis surface	Stores and matures sperm
Vas deferens	Long tube rising into the abdomen	Pushes sperm towards the urethra
Ejaculatory duct	Where vas deferens meets seminal vesicle	Delivers sperm plus fluid into the urethra
Urethra	Single shared exit tube	Carries urine or semen, never both together
Seminal vesicles	Paired glands behind the bladder	Add most of the semen fluid, rich in fructose for energy
Prostate gland	Single gland below the bladder	Adds milky alkaline fluid that protects sperm
Bulbourethral (Cowper's) glands	Two small pea-sized glands	Add clear mucus that lubricates and cleans the path

EVERYDAY EXAMPLE — A CAR FACTORY

Picture a car plant and map it honestly. The **seminiferous tubules** are the production floor where cars (sperm) are built. **Sertoli cells** are the line supervisors who feed parts to each worker. **Leydig cells** are the factory's announcement system — their message (testosterone) travels body-wide and tells every tissue to grow male features. The **epididymis** is the finished goods warehouse, the **vas deferens** the highway out, and the three glands are the paint-and-fuel stations that finally hand over a working vehicle (semen).

WORKED EXAMPLE — TRACE THE JOURNEY

A question lists six stops in scrambled order: urethra, epididymis, rete testis, vas deferens, vasa efferentia, ejaculatory duct. Put them in the order a sperm travels.

- **Start where sperm are born:** the seminiferous tubules empty into the **rete testis**.
- **Small pipes next:** the **vasa efferentia** lift sperm out to the epididymis.
- **Warehouse, then highway:** sperm mature in the **epididymis**, then ride the **vas deferens** up and over the bladder.
- **Final blend:** the **ejaculatory duct** joins the urethra, and the **urethra** carries semen out.

Answer: rete testis → vasa efferentia → epididymis → vas deferens → ejaculatory duct → urethra. Learn the ladder once; sequence questions become free marks.

The Female Reproductive System

The female system fits in a space smaller than your hand, yet it runs a monthly construction project, catches a tiny egg, and grows a whole baby when needed. The ovaries make the eggs and the hormones; a pair of tubes carries each egg a few centimetres; and the uterus is the room that gets redecorated every month. Here too, the game is the same: part → structure → function.

- **The ovaries — paired gonads:** almond-shaped glands, one on each side of the abdomen, that make eggs (ova) plus the hormones oestrogen and progesterone.
- **Oestrogen:** the hormone that develops female traits and rebuilds the uterine lining; progesterone keeps that lining in place.
- **Fallopian tubes (oviducts):** each is about 10–12 cm long and has three parts in a fixed order: infundibulum → ampulla → isthmus.
- **Infundibulum:** the funnel-shaped end near the ovary, fringed with finger-like **fimbriae** that sweep the released egg into the tube.
- **Ampulla:** the wider middle stretch — the usual site where sperm meets egg.
- **Isthmus:** the narrow last stretch that opens into the uterus.
- **The uterus — the womb:** an upside-down pear, about the size of a fist, with three wall layers (next table).
- **Cervix:** the narrow neck of the uterus that opens into the vagina.
- **Vagina:** the passage that receives semen, lets the baby out at birth, and carries the monthly flow out.
- **External genitalia:** together called the vulva — skin folds and openings around the vaginal entrance.
- **Mammary glands:** the milk glands of the breasts; tiny sacs called alveoli make the milk, and ducts carry it to the nipple.

Every part, one line each

Part	Structure in one line	The job it does
Ovary	Almond-sized paired gland	Makes eggs; releases oestrogen and progesterone
Fimbriae	Finger-like fringes on the tube end	Catch the egg released into the body cavity
Infundibulum	Funnel of the fallopian tube	Guides the egg into the tube
Ampulla	Wider middle part of the tube	Usual site of fertilisation
Isthmus	Narrow last part of the tube	Passes the egg on to the uterus
Perimetrium	Thin outer covering of the uterus	Wraps and protects
Myometrium	Thick middle muscle layer	Contracts strongly during labour

Part	Structure in one line	The job it does
Endometrium	Inner gland-rich lining	Rebuilds every month; the bed an embryo implants in; sheds if unused
Cervix	Narrow neck of the uterus	Opens into the vagina; widens at birth
Vagina	Muscular tube to the outside	Receives semen; birth canal; exit for the monthly flow
Mammary alveoli	Tiny milk sacs in the breast	Make and store milk
Milk ducts	Branching tubes from alveoli	Carry milk to the nipple

EVERYDAY EXAMPLE — A CATCH AT THE BOUNDARY

The ovary is not connected to the fallopian tube by a pipe — the egg is released straight into the small body space around it. The **fimbriae** are the fielder at the boundary: they flex and sweep just as the egg pops out, and their motion pulls it into the **infundibulum**. If the catch is missed, the egg has no road to the uterus — that is why the fimbriae moment matters.

WORKED EXAMPLE — WHICH WALL LAYER DOES WHAT?

Three statements, three layers — decide which layer each statement names.

- **"I contract and push the baby out."** Muscle that contracts = **myometrium**.
- **"I am rebuilt and shed every month."** The monthly lining = **endometrium**.
- **"I am just the thin outer wrap."** Outer cover = **perimetrium**.

Trap to note: students often say the **myometrium** sheds. Only the endometrium bleeds and sheds — the muscle layer stays put.

Quick drill — one line each

Question	Answer in one line
First part of the tube to receive the egg?	Infundibulum — the fimbriae sweep the egg in
Usual fertilisation site?	The ampulla — the tube's wider middle stretch
Which wall layer is muscle for labour?	Myometrium
Which wall layer rebuilds and sheds monthly?	Endometrium
Milk makers vs milk carriers?	Alveoli make milk; ducts carry it to the nipple

- **One egg a month:** after puberty the ovary releases one egg per cycle — and remember, the ovary's full stock of eggs was laid down before birth.

- **Two-way traffic:** the tube carries the egg downward while sperm swim upward past the isthmus and ampulla — the two meet in the middle stretch.

Spermatogenesis — the Sperm Factory

Spermatogenesis is the name for the whole process that turns a simple lining cell into a swimming sperm. It is a story of two cell divisions in a fixed order, and the classic check here is simple: how many cells, and with how many chromosomes, at each stage? Keep counting as you read — 46 halving to 23 is the whole trick. The process starts only at **puberty**, when hormones switch it on.

- **Definition in one line:** spermatogenesis = the making of sperm (spermatozoa) from male germ cells inside the seminiferous tubules.
- **Step 1 — spermatogonia multiply:** each spermatogonium (46 chromosomes) divides by **mitosis**; some daughters stay as reserve stock, others grow.
- **Step 2 — primary spermatocyte:** a grown-up spermatogonium (still 46) is now called a primary spermatocyte, ready for the first meiotic division.
- **Step 3 — meiosis-I:** the primary spermatocyte (46) splits into **two secondary spermatocytes**, each with 23 chromosomes.
- **Step 4 — meiosis-II:** each secondary spermatocyte splits again, giving **two spermatids** — so four spermatids from one primary spermatocyte, all with 23.
- **Step 5 — spermiogenesis:** each round spermatid reshapes into a slim, swimming **spermatozoon** (sperm). No cell division here — only remodelling.
- **Step 6 — spermiation:** the Sertoli cells release the finished sperm into the tubule's central canal (the lumen).

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THE CELL LADDER — ONE LINE AT A TIME

1 spermatogonium (46) → mitosis → 1 primary spermatocyte (46)

1 primary spermatocyte (46) → meiosis-I → 2 secondary spermatocytes (23 each)

1 secondary spermatocyte (23) → meiosis-II → 2 spermatids (23 each)

4 spermatids (23) → spermiogenesis → 4 spermatozoa (23)

Carry the bottom line: one primary spermatocyte gives **4 sperm** — equal, tiny and motile. Contrast with oogenesis in the next section.

Hormones that run the factory

- **GnRH starts it all:** the hypothalamus releases GnRH (gonadotropin-releasing hormone), which reaches the pituitary and switches it on.
- **LH (ICSH) — the testosterone switch:** luteinising hormone acts on Leydig cells, which secrete testosterone; without it, sperm cannot be made.
- **FSH — the nurse signal:** follicle-stimulating hormone acts on Sertoli cells, which then support and release the growing sperm.
- **Both together:** testosterone + FSH push spermatogenesis forward — a classic matching question pairs each hormone with the cell it acts on.

Reading a sperm

Part	What it holds	Why it matters
Head	The nucleus with 23 chromosomes, capped by the acrosome	Carries the father's 23 chromosomes to the egg
Acrosome	A cap over the head, packed with enzymes	Dissolves a path through the egg's protective layers
Middle piece	Many mitochondria (cell powerhouses) coiled tightly	Burns fuel to whip the tail — the sperm's engine
Tail	A long flagellum (whip-like propeller)	Pushes the sperm forward through the fluids

- **Size check:** a human sperm is about 60 micrometres long — a head, an engine and a propeller packed into one cell.

The hormone chain, cell by cell

Signal	Made by	Acts on	Effect
GnRH	Hypothalamus	Pituitary	Switches LH and FSH release on
LH (ICSH)	Pituitary	Leydig cells	They secrete testosterone
FSH	Pituitary	Sertoli cells	They feed sperm and release finished ones
Testosterone	Leydig cells	Tubules and the whole body	Builds male traits; supports sperm making

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- **Why the chain matters:** for any "which hormone acts where" question, name the **cell** first — hormone-to-cell matching is the whole trick.
- **Numbers to carry:** 46 halving to 23, four sperm per primary spermatocyte, sperm about 60 micrometres long, testes running 2–2.5 °C cooler.

WORKED EXAMPLE — SIX JUDGES, TRUE OR FALSE

Judge each statement before you read the verdict.

- **"Spermatogonia divide by meiosis."** — **False.** They divide by mitosis; meiosis begins with the primary spermatocyte.
- **"One primary spermatocyte gives four sperm."** — **True.** Two meiotic divisions, four equal products.
- **"Spermiogenesis is a cell division."** — **False.** It is pure reshaping of a spermatid into a sperm; no division happens.
- **"Spermiation and spermiogenesis are the same."** — **False.** Spermiogenesis reshapes the cell; spermiation is the Sertoli cell releasing it into the lumen.

- **"FSH acts on Leydig cells."** — **False.** FSH acts on Sertoli cells; LH (ICSH) acts on Leydig cells.
- **"Sperm are made at body temperature."** — **False.** They need 2–2.5 °C below body temperature, which is the scrotum's whole point.

Oogenesis — the Egg Factory

Oogenesis makes eggs, and it breaks every rule you just learned. It starts **before the girl is born**, pauses twice, and ends with one big egg plus tiny discarded cells instead of four equal sperm. The pauses are the most-asked facts in this chapter, so we will name each one and the exact stage it freezes at.

- **Before birth — the big start:** in the foetal ovary, oogonia (46) multiply by mitosis, then enter meiosis-I and **freeze in prophase-I** as primary oocytes.
- **Key exam fact:** every primary oocyte is already arrested (paused) in prophase-I **before the girl is even born** — no new oogonia are made after birth.
- **At puberty — one a month:** each month, one primary oocyte finishes meiosis-I.
- **Meiosis-I is unequal:** the cell splits into a huge **secondary oocyte** (23 chromosomes) plus a tiny **first polar body** — a half-sized cell that gets discarded.
- **Second pause:** the secondary oocyte freezes in **metaphase-II** and is released at ovulation in that state.
- **Meiosis-II finishes only after sperm entry:** when a sperm enters, the secondary oocyte completes meiosis-II, giving the **ovum** (23) plus a second polar body.
- **The follicle around the egg:** follicle cells surround and feed the oocyte; the mature, fluid-filled one is the **Graafian follicle**, and its burst is ovulation.
- **Zona pellucida:** a thick glycoprotein jacket (sugar-protein layer) around the egg that sperm must cross.
- **After the burst:** the empty follicle turns into the **corpus luteum**, which secretes progesterone — the pregnancy-support hormone.

Spermatogenesis vs Oogenesis — side by side

Question	Spermatogenesis	Oogenesis
Where?	Seminiferous tubules of the testis	Follicles of the ovary
When does it start?	Only at puberty	Before birth (arrested till puberty)
Cells from one parent cell	4 equal sperm	1 ovum + polar bodies (discarded)
Pauses (arrests)?	None — runs straight through	Twice: prophase-I before birth; metaphase-II till sperm entry

Question	Spermatogenesis	Oogenesis
Parent cell size	Equal, tiny divisions	Very unequal — almost all cytoplasm goes to one egg
Hormone that runs it	FSH on Sertoli; LH/ICSH on Leydig	FSH on follicles; LH surge bursts the Graafian follicle
Finished product	Motile sperm, made continuously	One egg per month, non-motile, food-rich

EVERYDAY EXAMPLE — TWO DIVISIONS, TWO STYLES

Spermatogenesis is a four-cup idli maker: one batch of batter (46) is poured so that four equal, light idlis (23 each) come out at once. **Oogenesis** is a single-bowl celebration cake: almost all the batter (the cytoplasm and food store) goes into one big cake (the ovum), while two-three tiny crumbs (the polar bodies) are swept off the counter. The mapping is exact — equal share versus near-total share to one product.

WORKED EXAMPLE — COUNT THE CELLS

State the chromosome number and cell count at each point.

- **One primary spermatocyte after meiosis-I:** two secondary spermatocytes, each with **23** chromosomes.
- **Both secondary spermatocytes after meiosis-II:** four spermatids, 23 each — so $2 \times 2 = 4$ **sperm** finally.
- **One primary oocyte after meiosis-I:** one secondary oocyte + one first polar body, both with **23** chromosomes — but only **1** will become an egg.

The same divisions, opposite products: 4 equal sperm versus 1 egg plus leftovers. That single line answers most gametogenesis questions.

The follicle grows with the egg

The egg never travels alone — a coat of follicle cells grows around it, stage by stage, and each stage has a name the exam checks.

Stage	What happens
Primary follicle	A single layer of cells wraps the primary oocyte
Secondary follicle	More layers of cells pile on around the oocyte
Tertiary follicle	A fluid-filled cavity appears; the oocyte sits off-centre
Graafian follicle	Mature and fluid-swollen — ready to burst on day 14
After the burst	The emptied follicle becomes the corpus luteum, the progesterone factory

- **Zona pellucida timing:** the thick glycoprotein jacket forms as the follicle grows — the sperm will have to cross it later.
- **Two stories, two halves:** before day 14 the follicle's story is oestrogen; after day 14 the corpus luteum's story is progesterone.

The Menstrual Cycle

Think of the cycle as the uterus preparing a bed every month. The ovary grows an egg, and the uterine lining turns into a soft, blood-rich bed for it. If the egg is not fertilised, the bed is stripped and rebuilt — and that shedding is the period. One cycle is counted from the first day of bleeding to the first day of the next, and the standard model length is **28 days**.

- **Four phases in order:** menstrual (days 1–5) → follicular (days 5–13) → ovulation (day 14) → luteal (days 15–28).
- **Menstrual phase (days 1–5):** the endometrium sheds out as the flow; all hormones sit at their lowest.
- **Follicular phase (days 5–13):** FSH grows several follicles in the ovary; one becomes dominant; its oestrogen rebuilds the endometrium.
- **Ovulation (day 14):** a sudden **LH surge** bursts the Graafian follicle; the secondary oocyte is released and caught by the fimbriae.
- **Luteal phase (days 15–28):** the empty follicle becomes the corpus luteum and makes progesterone, which keeps the bed warm and waiting.
- **If no fertilisation:** the corpus luteum dies after about 14 days, progesterone falls, the lining loses support — and day 1 begins again.
- **If fertilisation happens:** the corpus luteum is kept alive and progesterone keeps flowing, so no period comes and the pregnancy holds.

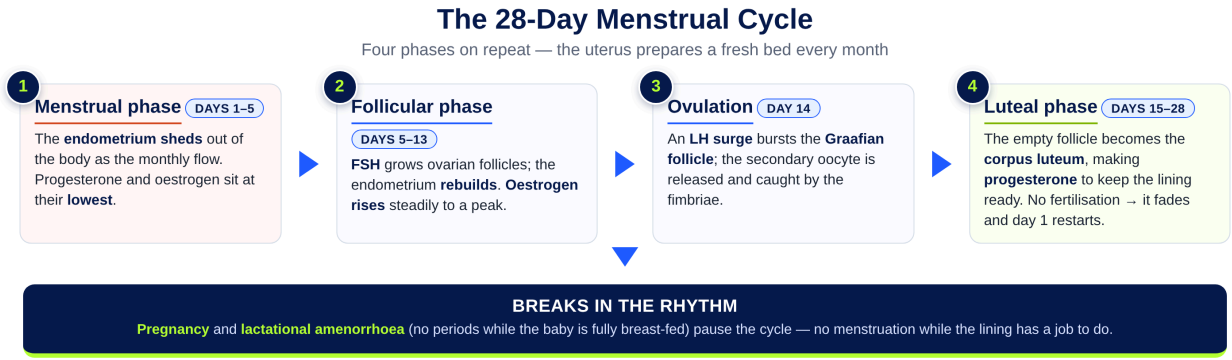


Figure 1: The 28-day menstrual cycle — four phases, their day ranges and the hormones that drive each one.

The hormone timeline, phase by phase

Phase	Days	Hormone at its peak	What the body does
Menstrual	1–5	All hormones low	Endometrium sheds as the monthly flow

Phase	Days	Hormone at its peak	What the body does
Follicular	5–13	FSH rising, then oestrogen climbing	Follicles grow; endometrium rebuilds layer by layer
Ovulation	14	LH surge (biggest spike of the month)	Graafian follicle bursts; secondary oocyte released
Luteal	15–28	Progesterone high, oestrogen moderate	Corpus luteum forms; lining stays plush and waiting

EVERYDAY EXAMPLE — THE MONTHLY HOTEL ROOM

Treat the uterus as a hotel room prepared for a guest. The **follicular phase** is housekeeping: fresh sheets (the endometrium) laid day by day. The **LH surge** is the doorbell — the egg arrives on day 14. **Progesterone** is the "room reserved" sign that keeps the sheets on. If the guest (an embryo) checks in, the sign stays up; if nobody arrives by day 28, housekeeping strips the room and the cycle restarts.

- **When the cycle pauses:** pregnancy stops it completely (no shed bed while an embryo rests on it).
- **Lactational amenorrhoea:** during intense breast-feeding, cycles often stay switched off — this is called lactational amenorrhoea (amenorrhoea = absence of menstruation).
- **Menarche — the first period:** the cycle's starting whistle at puberty, when the hormone system first switches the whole routine on.
- **Menopause — the last period:** at about 45–50 years of age the ovaries retire, cycles stop for good, and menstruation ends.

Two keywords per phase

Phase (days)	Ovary keyword	Uterus keyword
Menstrual (1–5)	All hormones low	Lining sheds as the flow
Follicular (5–13)	FSH grows follicles; oestrogen rises	Lining rebuilds layer by layer
Ovulation (14)	LH surge bursts the Graafian follicle	Egg released and caught by fimbriae
Luteal (15–28)	Corpus luteum makes progesterone	Lining stays plush and waiting

- **Revision trick:** say the two keywords aloud in order — "shed, rebuild, burst, wait" — and the 28-day map replays itself in four words.

Fertilisation, Cleavage and Implantation

One sperm, one egg, and a race with a deadline. Of the millions of sperm released, only a few hundred reach the tube, and just one is allowed inside. From that single meeting, the body runs a fixed programme: divide, travel, bury. Learn the three numbers — day of meeting, day of arrival, day of burial — and this section becomes easy marks.

- **The deposit:** semen is released into the vagina (about 200–300 million sperm per release); sperm swim up through the uterus into the fallopian tubes.
- **Capacitation:** the fluids of the female tract prime the sperm's outer membrane — only then can it fertilise.
- **The acrosome reaction:** at the egg, the acrosome bursts and releases enzymes that digest a path through the cells and the zona pellucida.
- **The cortical reaction:** the moment one sperm enters, the egg's outer layer hardens so no second sperm can enter — the polyspermy block.
- **Where it happens:** fertilisation occurs at the **ampullary-isthmic junction** — the meeting point of the tube's middle and last parts.
- **Sperm entry completes meiosis-II:** the secondary oocyte finishes its second division, giving the ovum (23) plus a second polar body.
- **The zygote:** sperm nucleus (23) + ovum (23) = zygote with 46 chromosomes — the single cell that starts a whole human.

Cleavage: divide first, grow later

- **Cleavage:** a series of quick mitotic divisions of the zygote; the cells divide again and again **without growing in between**.
- **The count:** 2 → 4 → 8 → 16 cells; the 16-cell ball is the **morula** (Latin for mulberry — it looks like one).
- **The journey:** cleavage starts in the tube while cilia (tiny hairs) and muscle push the morula towards the uterus.
- **The blastocyst:** in the uterus the morula becomes a hollow ball — outer flat cells (trophoblast) + a knob of cells inside (inner cell mass).
- **Who becomes what:** the trophoblast grows into the placenta's embryonic side; the inner cell mass becomes the embryo proper.
- **Implantation:** about 6–7 days after fertilisation, the trophoblast attaches to the endometrium, and uterine cells grow over and bury it.

Stage	Cells	Where it sits
Zygote	1 (46 chromosomes)	Ampullary-isthmic junction of the tube
Early cleavage (2, 4, 8 cells)	2–8	Still travelling down the fallopian tube
Morula	16	Near the end of the tube
Blastocyst	About 100+	Free in the uterine cavity

Stage	Cells	Where it sits
Implanted blastocyst	Trophoblast + inner cell mass	Buried in the endometrium (6–7 days after fertilisation)

EVERYDAY EXAMPLE — THE GUEST CHECKS IN

Recall the hotel from the last section. The egg was the doorbell day; now the guest actually arrives — a blastocyst drifting into a prepared room. The **trophoblast** is the guest's sticky luggage: it presses against the sheets (the endometrium) and the room wraps around it (implantation). From then on, room service runs day and night for nine months — and that service desk is the placenta, next section.

The barriers one sperm must cross

Barrier	How it is crossed
Fluids of the female tract	Capacitation primes the sperm's outer membrane
Cells around the egg	Acrosomal enzymes push a path through
Zona pellucida	Enzymes digest a way in; exactly one sperm enters
The egg's hardening shell	The cortical reaction blocks every second sperm — the polyspermy block

- **Why the block matters:** two sperm entering one egg would give 69 chromosomes — the cortical reaction keeps the count at exactly 46.
- **Symmetry of the meeting:** the sperm arrives with 23 and finishes the egg's meiosis-II, so both halves contribute 23 — zygote 46.
- **Division drill:** from 1 zygote to a 16-cell morula takes exactly **four rounds** of cleavage — $1 \rightarrow 2 \rightarrow 4 \rightarrow 8 \rightarrow 16$.
- **Fluid in the ball:** the blastocyst carries a fluid-filled space that helps it swell and settle onto the lining before implantation.

WORKED EXAMPLE — THE NUMBER JOURNEY

Follow the numbers from first meeting to buried blastocyst.

- **At fertilisation:** sperm 23 + ovum 23 = **zygote 46 chromosomes**.
- **By cleavage's end:** 16 cells — the morula, each cell still 46.
- **At the blastocyst:** about 100+ cells in two teams — trophoblast outside, inner cell mass inside.
- **At implantation:** roughly **6–7 days** after fertilisation, buried in the endometrium.

Four numbers — 46, 16, two teams, day 6–7 — carry this whole section.

Pregnancy, Placenta and Development

From implantation onwards, mother and embryo share one traffic junction: the placenta. It is the baby's lungs, kitchen and dustbin rolled into one disc, while the baby builds itself organ by organ on a fixed month-wise timetable. Doctors know that timetable by heart, and so should you — the milestone table below is a direct question bank.

- **How the placenta forms:** after implantation, finger-like **chorionic villi** grow from the trophoblast into the uterine tissue; the interlocking of villi and uterine tissue is the placenta.
- **Nutrition:** glucose, amino acids (protein building blocks) and vitamins cross from mother's blood to the foetus.
- **Respiration:** the placenta is the foetus's lungs — oxygen crosses in, carbon-dioxide crosses out.
- **Excretion:** the foetus's nitrogen wastes (like urea) cross into the mother's blood, which filters them out.
- **The umbilical cord:** the rope connecting foetus to placenta, carrying two arteries and one vein.
- **Placental hormones:** human chorionic gonadotropin (hCG), human placental lactogen (hPL), oestrogens and progesterone.
- **Relaxin:** an ovarian hormone of late pregnancy that relaxes the pelvic ligaments and softens the cervix for birth.

Foetal milestones — the month-wise timetable

Time after fertilisation	What the foetus can do
By the end of the first month	The heart is formed and beating
By the end of the second month	Limbs and digits (fingers and toes) appear
By the end of 12 weeks	Most major organ systems are laid down (first trimester done)
By the end of the fifth month	First movements and hair on the head show
By the end of 24 weeks	Eyelids separate and eyelashes appear
By the end of nine months	The body is fully formed and properly proportioned

- **Three layers, one baby:** the early embryo's cells sort into ectoderm (outer layer), mesoderm (middle) and endoderm (inner); every adult organ system traces back to one of these three germ layers.
- **First trimester = building,** later months = growing and maturing — that is why the first twelve weeks carry the highest risk if harmful drugs or infections cross the placenta.

WORKED EXAMPLE — MILESTONE MATCH

Match each event to its timing, then check yourself against the table.

- **"Heart is formed and beating"** → by the end of the **first month** — the earliest milestone in the list.
- **"Limbs and digits"** → by the end of the **second month**.
- **"Eyelids separate, eyelashes appear"** → by the end of **24 weeks** — sixth month, not the fifth.
- **"Body fully proportioned"** → end of the **ninth month**.

Trap: fifth month = movements + hair; sixth month (24 weeks) = eyelids + eyelashes. Students shift these by one month.

Three germ layers, every organ

Germ layer	What it grows into
Ectoderm (outer)	Skin surface and the whole nervous system
Mesoderm (middle)	Muscles, bones, heart, blood vessels, kidneys
Endoderm (inner)	Gut lining, lungs, liver, pancreas

- **Nine months, three trimesters:** pregnancy runs about nine months, counted as three trimesters of three months each.
- **Hormone handover:** the placenta's own hormone output rises through the first trimester, alongside the corpus luteum's progesterone supply.
- **Cushion and shock-absorber:** the foetus floats in amniotic fluid, which protects it from bumps and lets it move freely.
- **The pregnancy-test hormone:** hCG appears in the mother's urine in early pregnancy — home pregnancy kits detect exactly this hormone.
- **One rope, three vessels:** the umbilical cord carries two arteries and one vein between foetus and placenta.

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WORKED EXAMPLE — WHO MAKES WHICH PREGNANCY HORMONE?

Match each hormone to its maker and its job.

- **hCG** → placenta; keeps the corpus luteum alive in early pregnancy, so progesterone never dips.
- **hPL** → placenta; helps meet the mother's energy demands of pregnancy.
- **Oestrogens + progesterone** → placenta (the big late-pregnancy supply); progesterone holds the lining, oestrogens grow the uterus.
- **Relaxin** → **ovary**, not the placenta — it relaxes pelvic ligaments and softens the cervix for the birth to come.

Trap: relaxin is the odd one out — students file it under the placenta.

Parturition and Lactation

Birth is not an accident on a random day — it is a reflex with a schedule. When the baby is fully developed, signals pass between foetus, placenta and mother, mild contractions begin, and a hormone loop takes over that grows stronger until the baby is out. Then the same body switches on its next job: feeding the newborn.

- **Parturition:** the technical name for the delivery of the baby (birth).
- **The signal to start:** the fully developed foetus and the placenta send signals that trigger mild uterine contractions — the **foetal ejection reflex**.
- **Foetal cortisol in one line:** the mature foetus releases cortisol (a stress hormone), one of the signals that helps switch labour on.
- **Oxytocin enters:** the contractions signal the mother's posterior pituitary to release **oxytocin**.
- **The loop:** oxytocin makes contractions **stronger** → stronger contractions make the pituitary release **more** oxytocin → the loop repeats until the baby is expelled.
- **This is positive feedback:** the output (contractions) increases its own cause (oxytocin) — the opposite of the calming negative feedback you meet in temperature control.
- **Stages of birth:** labour (cervix widens) → baby delivered → **afterbirth** (the placenta is expelled last).

Stage	What happens
Labour	Strong rhythmic contractions widen the cervix
Delivery	The baby is pushed out through the birth canal
Afterbirth	The placenta detaches and is expelled last

Lactation: the milk switch

- **After delivery:** high prolactin turns the mammary glands on; the alveoli start making milk.
- **Oxytocin's second job:** it squeezes the milk ducts to push milk out when the baby suckles (the let-down reflex).
- **Colostrum:** the **first milk** — thick and yellowish, made in the first few days.
- **Why colostrum matters:** it is packed with **IgA antibodies** (proteins that fight infection), which coat the newborn's gut — the baby's first immunity before its own immune system matures.

EVERYDAY EXAMPLE — THE GROWING APPLAUSE

Positive feedback is a crowd's applause loop: a few people clap (mild contractions), the noise excites more people to join (more oxytocin), the clapping grows louder (stronger contractions), which excites even more people — until the hall erupts (the baby is out). In negative feedback the crowd would **hush itself down**; in birth, the loop must build, so the applause only grows. The mapping is honest: each round strengthens the next until the event completes.

WORKED EXAMPLE — PUT THE BIRTH SEQUENCE IN ORDER

A question scrambles five events: baby delivered, placenta expelled, mild contractions begin, oxytocin loop strengthens, cervix widens.

- **Step 1:** foetal + placental signals → mild contractions begin.
- **Step 2:** posterior pituitary releases oxytocin; the loop strengthens contractions.
- **Step 3:** cervix widens through labour.
- **Step 4:** baby delivered.
- **Step 5:** afterbirth — placenta expelled.

The loop sits **before** delivery, and the placenta always leaves **last** — both classic sequence checks.

The hormone cast at the end

Hormone	Source	Job here
Oxytocin	Posterior pituitary	Strengthens contractions in the birth loop; later pushes milk out
Prolactin	Pituitary gland	Switches the mammary glands on after delivery
Relaxin	Ovary	Relaxes pelvic ligaments; softens the cervix

- **Two jobs, one hormone:** oxytocin runs birth and breast-feeding — stronger contractions before the birth, milk let-down after it.
- **Feeding keeps feeding:** suckling keeps both hormones working — more suckling, more milk made and released.
- **Why breast helps early:** the IgA from colostrum coats the baby's gut, which is why breast-fed newborns face fewer early gut infections.

Colostrum vs mature milk

Feature	Colostrum (first days)	Mature milk (after 1 week)
Look	Thick and yellowish	Thinner and whitish
Special load	Packed with IgA antibodies	Balanced food for growth
Job	Baby's first immunity shield	Steady fuel and building material

- **Same gland, two recipes:** the alveoli change their output over the first week — immunity first, volume next.

Key Facts & Terms Cheatsheet

KEY FACTS & TERMS CHEATSHEET — HUMAN REPRODUCTION

Term / fact	What it means	Example / number to carry
Scrotum temperature gap	Testes sit outside the body because sperm need cooler air	2–2.5 °C below body temperature
Sertoli cells	Nurse cells inside the tubule wall	Feed developing sperm; do spermiation (release)
Leydig cells	Interstitial cells between tubules	Secrete testosterone, driven by LH/ICSH
Duct ladder	Route sperm travel out of the testis	Rete testis → vasa efferentia → epididymis → vas deferens → ejaculatory duct → urethra
Tube parts in order	The three stretches of each fallopian tube	Infundibulum (fimbriae) → ampulla → isthmus
Uterus wall layers	Outer to inner	Perimetrium → myometrium (muscle) → endometrium (lining that sheds)
Spermatogenesis yield	Equal products from one parent cell	1 primary spermatocyte → 4 sperm (23 each)
Oogenesis yield	Very unequal products	1 primary oocyte → 1 ovum + polar bodies (discarded)
Arrest 1	Meiotic pause before birth	Primary oocyte frozen in prophase-I
Arrest 2	Meiotic pause at ovulation	Secondary oocyte frozen in metaphase-II until sperm entry
Cycle phases	The 28-day model	Menstrual 1–5 → follicular 5–13 → ovula-

Term / fact	What it means	Example / number to carry
		tion day 14 → luteal 15–28
Ovulation trigger	The day-14 hormonal event	LH surge bursts the Graafian follicle
Fertilisation site	Where sperm meets egg	Ampullary-isthmic junction
Morula vs blastocyst	The two early embryo stages	Morula = 16 cells; blastocyst = hollow ball (trophoblast + inner cell mass)
Implantation timing	When the blastocyst buries itself	6–7 days after fertilisation
Placenta hormones	Four from the placenta	hCG, hPL, oestrogens, progesterone (relaxin from the ovary)
Milestones	Month-wise foetal timetable	Heart month 1; limbs month 2; organ systems by 12 weeks; hair month 5; eyelids 24 weeks
Oxytocin loop	The birth reflex	Positive feedback: contractions → oxytocin → stronger contractions
Colostrum	The first milk	Yellow, thick, rich in IgA antibodies

One-line revision chain: cooler testes → tubules → 46 to 23 → 4 sperm or 1 egg → day-14 LH surge → ampulla meeting → 46 zygote → morula 16 → blastocyst → buried by day 7 → placenta takes over → oxytocin loop → colostrum first.

Common Errors and Special Cases

COMMON ERRORS & SPECIAL CASES

The mistake students make	What is right	Why / the trap
Saying meiosis in males starts before puberty	Spermatogonia divide by **mitosis** until puberty; meiosis starts when hormones switch on	Puberty is the meiosis gate
Treating oogenesis as one continuous run	It arrests **twice**: prophase-I before birth, metaphase-II until a sperm enters	Two pauses, two stages
Blaming the FSH surge for ovulation	The **LH surge** (day 14) bursts the Graafian follicle	FSH grows follicles; LH bursts them
Placing fertilisation in the uterus	It happens at the **ampullary-isthmic junction** of the tube	Uterus hosts implantation, not fertilisation
Counting polar bodies as eggs	Polar bodies are tiny, discarded leftovers	Only one usable product per oocyte
Swapping Sertoli and Leydig jobs	**Sertoli** nourish sperm; **Leydig** secrete testosterone	Nurse vs hormone-maker
Calling birth negative feedback	Oxytocin loop is **positive** feedback — output increases its own cause	The loop must build, not calm down

Three checking habits before you mark an option: (1) name the **stage** of every meiotic pause, not just "paused"; (2) for any hormone question, name the **cell** it acts on first; (3) before saying "uterus", ask whether the event is fertilisation (tube) or implantation and pregnancy (uterus).

The Whole Chapter on One Page

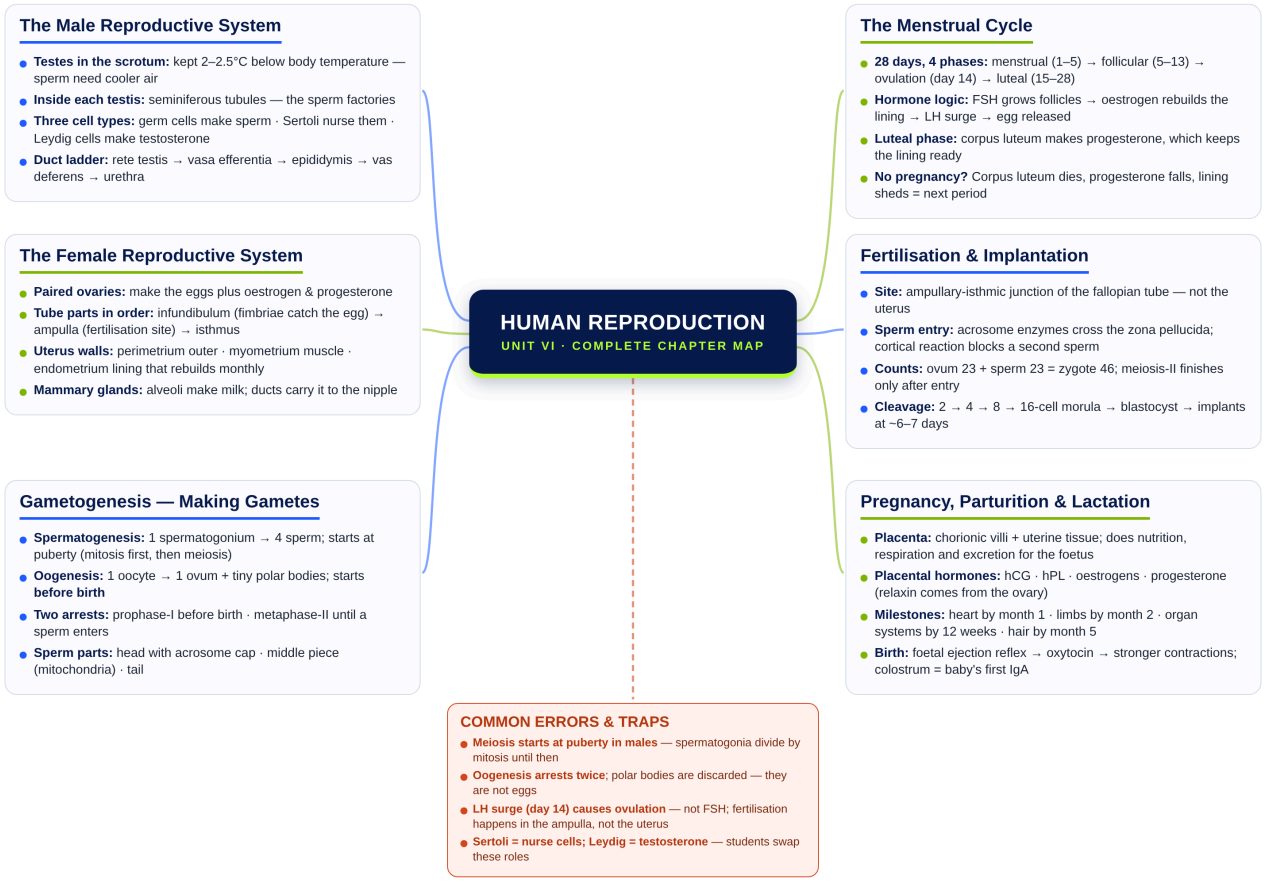


Figure 2: Chapter mindmap — the six blocks of Unit VI and the traps that catch most students.

Practice Questions

Q1. The testes are held in the scrotum, outside the abdominal cavity, because sperm formation needs a temperature that is:

(A) equal to normal body temperature
(B) 2–2.5 °C above body temperature
(C) 2–2.5 °C below body temperature
(D) 4–5 °C above body temperature

Q2. Which sequence correctly shows the path a sperm travels from the seminiferous tubules to the outside?

(A) epididymis → rete testis → vasa efferentia → vas deferens → urethra
(B) rete testis → epididymis → vasa efferentia → vas deferens → urethra
(C) vasa efferentia → rete testis → epididymis → vas deferens → urethra
(D) rete testis → vasa efferentia → epididymis → vas deferens → ejaculatory duct → urethra

Q3. Testosterone, the hormone that drives sperm production and male traits, is secreted by:

(A) interstitial (Leydig) cells
(B) Sertoli cells
(C) spermatogonia
(D) secondary spermatocytes

Q4. A primary spermatocyte with 46 chromosomes finally yields how many sperm, each with how many chromosomes?

(A) 2 sperm with 46 chromosomes each
(B) 4 sperm with 23 chromosomes each
(C) 4 sperm with 46 chromosomes each
(D) 1 sperm with 23 chromosomes

Q5. In a newborn girl, every primary oocyte is already paused at which stage of meiosis?

(A) metaphase-II
(B) anaphase-I
(C) prophase-I
(D) metaphase-I

Q6. The sudden hormonal event on day 14 of the cycle that bursts the Graafian follicle and releases the oocyte is the:

(A) FSH surge
(B) progesterone peak
(C) sudden fall of oestrogen
(D) LH surge

Q7. Sperm normally meets the egg at the:

(A) ampullary-isthmic junction of the fallopian tube
(B) uterine cavity
(C) cervix
(D) ovary

Q8. A blastocyst buries itself in the endometrium (implantation) about how many days after fertilisation?

(A) 1–2 days
(B) 6–7 days
(C) 14 days
(D) 28 days

Q9. Which set of hormones is produced by the placenta during pregnancy?

(A) FSH, LH, GnRH
(B) oxytocin, vasopressin, prolactin
(C) hCG, hPL, oestrogens, progesterone
(D) insulin, cortisol, adrenaline

Q10. Colostrum, the first milk, is precious for the newborn mainly because it is rich in:

(A) fat for rapid weight gain
(B) IgA antibodies that give first immunity
(C) starch for energy
(D) urea and salts

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Solutions

Sol. Q1. Sperm formation runs best slightly cooler than the body core, so the scrotum holds the testes **outside** at 2–2.5 °C below body temperature. Options above or equal to body temperature would damage spermatogenesis.

Answer: (C)

Sol. Q2. Trace the ladder: tubules empty into the **rete testis**, small **vasa efferentia** lift sperm to the **epididymis** (storage + maturation), the **vas deferens** pushes them up, the **ejaculatory duct** joins the **urethra**, and the urethra exits. Only option (D) keeps every rung in order.

Answer: (D)

Sol. Q3. LH (also called ICSH in males) acts on the **Leydig cells** sitting between the tubules, and they secrete testosterone. Sertoli cells are nurse cells; spermatogonia and secondary spermatocytes are germ cells, not hormone-makers.

Answer: (A)

Sol. Q4. Count it: one primary spermatocyte (46) → meiosis-I → two secondary spermatocytes (23 each) → meiosis-II → four spermatids (23) → spermiogenesis → **four sperm with 23 chromosomes each**. Equal, tiny products — the opposite of oogenesis.

Answer: (B)

Sol. Q5. Oogonia multiply before birth and enter meiosis-I, which freezes at **prophase-I**. The cell stays a primary oocyte until puberty; metaphase-II is the **second** arrest, reached only after meiosis-I completes at puberty.

Answer: (C)

Sol. Q6. The mid-cycle event is the **LH surge**: a sharp spike of luteinising hormone on about day 14 that ruptures the Graafian follicle. FSH grows follicles in the follicular phase but does not burst them.

Answer: (D)

Sol. Q7. The classic site is the **ampullary-isthmic junction** — where the tube's wide middle part (ampulla) meets its narrow last part (isthmus). The uterus is where implantation happens, not fertilisation.

Answer: (A)

Sol. Q8. After fertilisation, cleavage takes about 3–4 days in the tube, the blastocyst reaches the uterus, and implantation happens **about 6–7 days after fertilisation**. Fourteen and twenty-eight days belong to cycle timing, not burial.

Answer: (B)

Sol. Q9. The placenta is an endocrine organ: it secretes **hCG, hPL, oestrogens and progesterone**. FSH/LH and GnRH stay in the brain–pituitary axis, and oxytocin comes from the mother's posterior pituitary, not the placenta.

Answer: (C)

Sol. Q10. Colostrum is the thick, yellowish first milk of the first few days, loaded with **IgA antibodies** that coat the baby's gut and guard against infection while the baby's own immune system is still immature.

Answer: (D)

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